UNAM School of Engineering and the Built Environment
JEDS Campus | Semester Project Brief | Semester 1, 2026

Software Requirements Specification

CorroCheck

A Mobile Corrosion Inspection and Reporting System

Group Name	Group 7
Engineering Domain	Metallurgical Engineering
Institution	UNAM — School of Engineering and the Built Environment
Campus	JEDS Campus
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Section 1 — Introduction

1.1 Project Title, Group Name, and Team Members

Project Title: CorroCheck — A Mobile Corrosion Inspection and Reporting System

Group Name: Group 7

#	Full Name	Student Number	Role
1	Elise S Shaumbwa	225049473	Project Manager, Backend – Integration & Testing
2	Penny P Hamunyela	225058219	UI/UX Designer

#	Full Name	Student Number	Role
3	Nghoshi Immanuel	225062127	UI/UX Designer
4	Elia CN Daniel	222126698	UI/UX Designer
5	Emilia Alveta	222143738	UI/UX Designer
6	Touno E Ndivayele	224119346	UI/UX Designer
7	Martha N Martin	225076799	Frontend Developer
8	Tomas Shiyanga	225077663	Frontend Developer
9	Luiise N Shatilwapo	225060221	Frontend Developer
10	Ndateelela Nghuumono	225065150	Frontend Developer
11	Foibe N Hangula	222136855	Frontend Developer
12	Willhem N Hishongwa	225079224	Backend – Firebase Authentication
13	Natangwe Tuyapeni	225086034	Backend – Node.js & Roboflow API
14	Timoteus Johannes	225076985	Backend – Firebase Firestore
15	Elizabeth Nuugonya	225066963	Backend – Firebase Storage

1.2 Purpose of This Document

This Software Requirements Specification (SRS) formally defines the functional and non-functional requirements for the CorroCheck mobile application. It serves as the primary reference document against which the final system will be evaluated. Every feature implemented in Phase 4 must trace back to a numbered requirement contained in this document. This document is intended for use by the development team, the project manager, and the course lecturer (Mr. Abisai) as the agreed contract between the group and the client.

1.3 Scope — What the App Does and Does Not Do

What CorroCheck does:

- Allows registered users to capture or upload photographs of metallic structures for corrosion inspection.
- Provides AI-assisted corrosion severity classification (Low, Medium, High) via the Roboflow API.
- Allows users to select the type of corrosion observed from a predefined list.
- Captures GPS location automatically or accepts manual location input for each inspection.
- Stores all inspection reports securely in Firebase Firestore, linked to the authenticated user's account.
- Allows users to retrieve and review their full inspection history at any time.
- Supports user registration, login, and logout via Firebase Authentication.

What CorroCheck does NOT do:

- Does not perform laboratory-level metallurgical analysis or provide certified engineering assessments.
- Does not function as a replacement for professional corrosion engineering inspections.
- Does not support team or multi-user shared inspection environments in its initial version.
- Does not operate fully offline — an internet connection is required for cloud sync, image upload, and AI analysis.

1.4 Engineering Domain and Problem Statement

Engineering Domain: Metallurgical Engineering

Problem Statement:

Corrosion is one of the most costly and hazardous challenges in civil, mining, and metallurgical engineering. Metallic infrastructure — including pipelines, bridges, mining equipment, and structural supports — degrades over time due to chemical reactions with the environment. Traditional corrosion inspection methods rely on manual visual assessments conducted by trained engineers on-site, a process that is time-consuming, inconsistent across inspectors, and difficult to document systematically in the field.

There is currently no accessible, standardised mobile tool that allows engineers and field inspectors in Namibia to capture, classify, and store corrosion inspection data in real time. As a result, inspection records are often incomplete, delayed, or lost, making it difficult to monitor corrosion progression over time and take timely preventive action.

CorroCheck addresses this gap by providing a mobile application that enables engineers, inspectors, and maintenance personnel to conduct structured corrosion inspections directly from their smartphones — capturing photographic evidence, recording corrosion type and severity, logging the GPS location, and storing all data in a secure cloud database for future analysis and reporting.

Section 2 — System Overview

2.1 System Description

CorroCheck is a standalone mobile application built using React Native and Expo, targeting Android devices running API Level 29 (Android 10.0) and above. The application connects to Firebase for user authentication, cloud data storage, and image file storage, and integrates with the Roboflow API for AI-powered corrosion severity analysis. Users interact with the app through a simple, mobile-optimised interface designed to be usable by engineers and inspectors with no prior technical training.

2.2 Target Users

Civil Engineers	Inspect bridges, drainage systems, and structural metalwork for corrosion damage.

Mining Engineers	Assess corrosion on underground pipes, support structures, and mining equipment.
Metallurgical Engineers	Monitor material degradation and document findings for materials analysis.
Field Inspectors	Conduct routine site inspections and generate structured reports in real time.
Maintenance Personnel	Record corrosion findings during maintenance rounds for follow-up action.

2.3 Technology Overview

Mobile Framework	React Native + Expo (cross-platform, Android-first)
Backend Server	Node.js (handles photo forwarding and Roboflow API communication)
Authentication	Firebase Authentication (email and password)
Database	Firebase Firestore (inspection reports and user data)
File Storage	Firebase Storage (uploaded inspection photographs)
AI Analysis	Roboflow API (pre-trained corrosion detection model)
Version Control	Git + GitHub (organisation repository — Group 7)
Design Tool	Figma (UI/UX wireframes and screen design)

2.4 Firebase Data Model

Collection: users

Field	Type	Description
userId	String (UID)	Auto-generated unique identifier from Firebase Auth.
name	String	The user's full name, captured at registration.
email	String	The user's email address used for authentication.
createdAt	Timestamp	The date and time the account was created.

Sub-collection: users / {userId} / scans

Field	Type	Description
scanId	String (Auto-ID)	Auto-generated unique ID for each inspection scan.
photoURL	String (URL)	Link to the uploaded image stored in Firebase Storage.
corrosionType	String	Uniform, Galvanic, Pitting, or Crevice.
severity	String	Low, Medium, or High — returned by the Roboflow API.

Field	Type	Description
location	String	GPS coordinates or manually entered location description.
date	Timestamp	Date and time the inspection was conducted.
notes	String	Optional free-text notes added by the inspector.
confidence	Number	AI confidence score (percentage) returned by Roboflow.

2.5 Assumptions and Constraints

- The application requires a stable internet connection for Firebase synchronisation, image upload, and AI analysis via the Roboflow API.
- The application targets Android devices running API Level 29 (Android 10.0) or higher.
- Users must have a valid email address to register and access the application.
- The AI severity analysis is provided by a third-party service (Roboflow) and is intended as a field guide — it does not constitute a certified engineering assessment.
- All inspection photographs must be clear and well-lit for accurate AI analysis results.
- Firebase Firestore and Firebase Storage are assumed to be available and accessible throughout usage.

Section 3 — Functional Requirements

Each requirement below has a unique ID, is specific and testable, and references the Firebase service it involves. A minimum of 10 functional requirements are included as required by the project brief.

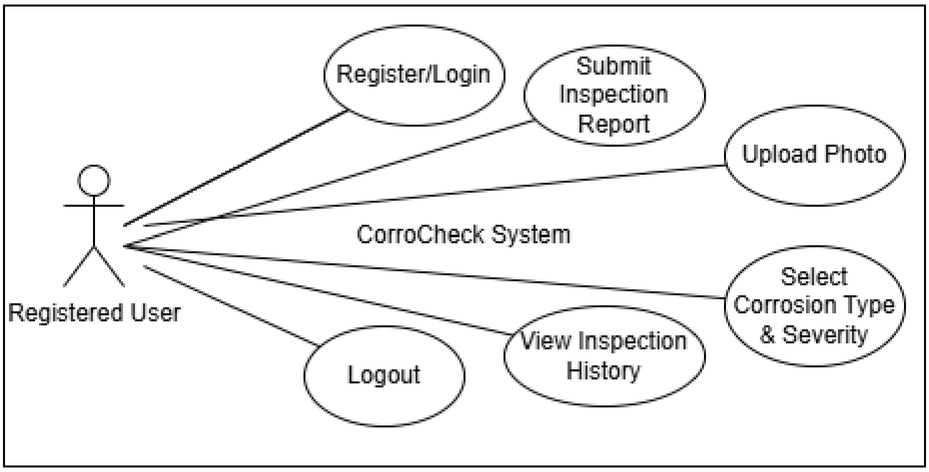
ID	Requirement Statement	Priority	Firebase Service
FR-001	The system shall allow a new user to register with a full name, email address, and password using Firebase Authentication.	High	Firebase Auth
FR-002	The system shall allow a registered user to log in and log out securely using their email address and password.	High	Firebase Auth
FR-003	The system shall allow an authenticated user to create a new inspection report by capturing or uploading a photograph.	High	Firebase Storage
FR-004	The system shall allow a user to select the corrosion type from the following options: Uniform, Galvanic, Pitting, or Crevice.	High	Firestore
FR-005	The system shall automatically capture the device's GPS coordinates when a new inspection report is created.	Medium	Device GPS
FR-006	The system shall allow a user to manually enter a location description if GPS is unavailable or disabled.	Medium	Firestore

ID	Requirement Statement	Priority	Firebase Service
FR-007	The system shall submit the uploaded photograph to the Roboflow API and return a severity classification of Low, Medium, or High.	High	Roboflow API / Node.js
FR-008	The system shall save the completed inspection report — including photo URL, corrosion type, severity, location, date, and notes — to Firestore under the authenticated user's account.	High	Firestore
FR-009	The system shall display all inspection reports submitted by the logged-in user in reverse chronological order on the History screen.	High	Firestore
FR-010	The system shall allow a user to tap any saved inspection report to view its full details, including the photograph, corrosion type, severity, location, date, and notes.	High	Firestore / Firebase Storage
FR-011	The system shall display the AI confidence score alongside the severity classification on the results screen.	Low	Roboflow API

Section 4 — Non-Functional Requirements

- Performance — the app shall load the main dashboard within 3 seconds on a standard Wi-Fi connection.
- Security — all Firestore rules shall require request.auth != null for any read or write operation, ensuring that unauthenticated users cannot access any data.
- Usability — a new user shall be able to complete registration and submit their first inspection report without external guidance.
- Compatibility — the app shall function correctly on Android 10.0 (API Level 29) and above.
- Reliability — locally entered data shall not be lost if the device briefly loses connectivity during report creation.
- Scalability — the Firebase Firestore data structure shall support multiple users simultaneously without degradation of performance.

Section 5 — Use Case Diagrams



Primary Actors

- Registered User — an engineer, inspector, or maintenance professional with an active CorroCheck account.
- Firebase Authentication — the external system responsible for verifying user identity.
- Firebase Firestore — the external system responsible for storing and retrieving inspection data.
- Roboflow API — the external AI service responsible for analysing corrosion severity from photographs.

Top Three Use Cases

Use Case 1 — Submit a Corrosion Inspection Report

A registered user opens the app and navigates to the inspection screen. They take or upload a photograph of a corroded metal surface, select the corrosion type, confirm the GPS location, and add optional notes. The app forwards the photo to the Roboflow API via the Node.js backend server, receives a severity classification, and saves the complete report to Firestore. The user is shown the result on screen immediately.

Use Case 2 — View Inspection History

A registered user navigates to the History screen. The app queries Firestore for all inspection reports linked to the user's account and displays them in reverse chronological order, showing a thumbnail, date, and severity badge for each entry. The user taps a report to view its full details, including the original photograph loaded from Firebase Storage.

Use Case 3 — Register and Log In

A new user opens the app for the first time and is directed to the registration screen. They enter their full name, email address, and password. Firebase Authentication creates their account and logs them in automatically. On subsequent visits, the user enters their email and password on the login screen and is authenticated by Firebase before being granted access to the dashboard.

Section 6 — Future Enhancements

The following features are outside the scope of the current version but are identified for consideration in future development phases:

- Integration of a custom-trained AI model fine-tuned specifically on Namibian industrial corrosion datasets for improved accuracy.
- PDF report export functionality, allowing users to generate and share formal inspection reports directly from the app.
- Smart alerts and push notifications to remind users of scheduled inspection intervals based on historical data.
- Team or organisation accounts allowing multiple inspectors to share and manage a collective inspection database.
- Offline mode with local data caching and automatic synchronisation when connectivity is restored.
- Dashboard analytics showing corrosion trends, frequency by location, and severity distribution over time.

Section 7 — Conclusion

CorroCheck provides a modern, accessible solution to the longstanding challenge of inconsistent and poorly documented corrosion inspection practices in civil, mining, and metallurgical engineering. By combining a mobile-first interface with cloud-based data management and AI-assisted severity classification, the application reduces manual errors, improves the consistency of inspection records, and ensures that all data is securely stored and immediately accessible to the inspector in the field.

This SRS defines the full scope of the system to be built during Phase 4. Every requirement listed herein is testable, traceable, and directly mapped to the Firebase services and technologies that will be used in implementation. This document is the contract between Group 7 and the course evaluation criteria, and all development decisions will be made in reference to it.

— Group 7, UNAM JEDS Campus, Semester 1, 2026